Skin Emotes (com.fffemotes.skintools.master) - Remote Config, Analytics & In-app update audit

Branch feature/ads-rc-infra, read from the code and docs/remote-config/*.json. Compared with the reference app 179-Body-Scanner-Home-01.

1. Summary

• Remote Config: every key in base and track_b is parsed AND used in app behaviour, except the top-level 'schema' version marker (nothing reads it). app_config_debug.json has the same keys.

• Paid value (ad revenue): every AdMob ad format sends Firebase 'ad_impression' with value + currency from the ad's OnPaidEvent (9 listeners).

• Log events: 12 fixed Firebase events (screen, click, key_press, permission, ad, ad_impression, default_home, app_update, rate_us, uninstall, install_source, first_home); every event carries 'segment'. Same event set as the reference app.

• In-app update: Google Play app-update 2.1.0, same as the reference; forced-update decline now closes the app (as the reference); versionName set to 1.0.0 like the reference.

Notes from the audit

Every leaf key in app_config.json base + track_b is parsed and has a real behavioural consumer, except top-level 'schema' (not parsed). app_config_debug.json has the identical key set. Configured-but-unreachable under the current RC values: ads.onboarding.welcome / permission_hub and permissions.permission_hub (steps absent from onboarding.order), launcher.host_app_panel / suggested_panel ads (panels disabled). launcher.home_button exit_ad only fires during onboarding. Parser defaults differ from JSON for ads.loader_on_miss (true vs false) and base.app_update.show (false vs true), relevant only if a key is ever dropped. Vault key googleS_Inter is written but unread (splash inter unit is taken directly from AdsConfig).

2. Paid value - ad_impression (ad revenue)

AdEvents.paid (ads/.../host/AdEvents.kt:44-56) turns every AdMob OnPaidEventListener into Firebase logEvent('ad_impression') = FirebaseAnalytics.Event.AD_IMPRESSION (Analytics.kt:20,53). It uses the GA4 standard ad_impression params ad_platform ('admob'), ad_source, ad_format (banner/native/interstitial/rewarded/app_open via standardFormat, AdEvents.kt:59), ad_unit_name, value (double, micros/1e6) and currency (falls back to 'USD'), plus custom ad_placement, ad_screen, user_segment, ad_place, segment. AdValue.precisionType is NOT logged. Covered formats: banner, native_banner (native_strip), all native slot types, full_native, interstitial, rewarded, app_open (exit step and resume). Every AdMob show path found has a paid listener. Strings are truncated to 100 chars (EventRules.clean); value stays a double so GA4 revenue reporting works.

Format	Where	Event	Parameters
banner (banner / banner_inline / banner_collapsible slots)	BannerRenderer.kt:189	ad_impression	ad_platform=admob, ad_source (loadedAdapterResponseInfo.adSourceName, else 'unknown'), ad_format (standardFormat(label)), ad_unit_name (unit id, else 'unknown'), value (valueMicros/1e6, double), currency (currencyCode, else 'USD'), ad_placement (engine label), ad_screen (Screens.current), user_segment (paid/organic), ad_place (RC place key; omitted when ''), segment (added to every event by Analytics.log)
native_banner slot	NativeStripRenderer.kt:175	ad_impression	ad_platform=admob, ad_source (loadedAdapterResponseInfo.adSourceName, else 'unknown'), ad_format (standardFormat(label)), ad_unit_name (unit id, else 'unknown'), value (valueMicros/1e6, double), currency (currencyCode, else 'USD'), ad_placement (engine label), ad_screen (Screens.current), user_segment (paid/organic), ad_place (RC place key; omitted when ''), segment (added to every event by Analytics.log)
native_big / native_mid / native_mid2 / native_full slots (cached ad shown)	NativeAdRenderer.kt:155 (labels from :266 native_big, :359 native_full, :624 native_medium, :766 native_medium2)	ad_impression	ad_platform=admob, ad_source (loadedAdapterResponseInfo.adSourceName, else 'unknown'), ad_format (standardFormat(label)), ad_unit_name (unit id, else 'unknown'), value (valueMicros/1e6, double), currency (currencyCode, else 'USD'), ad_placement (engine label), ad_screen (Screens.current), user_segment (paid/organic), ad_place (RC place key; omitted when ''), segment (added to every event by Analytics.log)
native loaded for a specific unit (presentWithUnit)	NativeAdRenderer.kt:197	ad_impression	ad_platform=admob, ad_source (loadedAdapterResponseInfo.adSourceName, else 'unknown'), ad_format (standardFormat(label)), ad_unit_name (unit id, else 'unknown'), value (valueMicros/1e6, double), currency (currencyCode, else 'USD'), ad_placement (engine label), ad_screen (Screens.current), user_segment (paid/organic), ad_place (RC place key; omitted when ''), segment (added to every event by Analytics.log)
full_native exit_ad step (full-screen native)	NativeAdRenderer.kt:441	ad_impression	ad_platform=admob, ad_source (loadedAdapterResponseInfo.adSourceName, else 'unknown'), ad_format (standardFormat(label)), ad_unit_name (unit id, else 'unknown'), value (valueMicros/1e6, double), currency (currencyCode, else 'USD'), ad_placement (engine label), ad_screen (Screens.current), user_segment (paid/organic), ad_place (RC place key; omitted when ''), segment (added to every event by Analytics.log)
inter exit_ad step (all interstitials, incl. splash and back unit)	ChainInterstitial.kt:173	ad_impression	ad_platform=admob, ad_source (loadedAdapterResponseInfo.adSourceName, else 'unknown'), ad_format (standardFormat(label)), ad_unit_name (unit id, else 'unknown'), value (valueMicros/1e6, double), currency (currencyCode, else 'USD'), ad_placement (engine label), ad_screen (Screens.current), user_segment (paid/organic), ad_place (RC place key; omitted when ''), segment (added to every event by Analytics.log)
rewarded exit_ad step	RewardAdRenderer.kt:101	ad_impression	ad_platform=admob, ad_source (loadedAdapterResponseInfo.adSourceName, else 'unknown'), ad_format (standardFormat(label)), ad_unit_name (unit id, else 'unknown'), value (valueMicros/1e6, double), currency (currencyCode, else 'USD'), ad_placement (engine label), ad_screen (Screens.current), user_segment (paid/organic), ad_place (RC place key; omitted when ''), segment (added to every event by Analytics.log)
app_open exit_ad step (launcher.app_open, uninstall_flow.splash, onboarding exits...)	ResumeAdDirector.kt:226	ad_impression	ad_platform=admob, ad_source (loadedAdapterResponseInfo.adSourceName, else 'unknown'), ad_format (standardFormat(label)), ad_unit_name (unit id, else 'unknown'), value (valueMicros/1e6, double), currency (currencyCode, else 'USD'), ad_placement (engine label), ad_screen (Screens.current), user_segment (paid/organic), ad_place (RC place key; omitted when ''), segment (added to every event by Analytics.log)
app_open resume path	ResumeAdDirector.kt:181	ad_impression	ad_platform=admob, ad_source (loadedAdapterResponseInfo.adSourceName, else 'unknown'), ad_format (standardFormat(label)), ad_unit_name (unit id, else 'unknown'), value (valueMicros/1e6, double), currency (currencyCode, else 'USD'), ad_placement (engine label), ad_screen (Screens.current), user_segment (paid/organic), ad_place (RC place key; omitted when ''), segment (added to every event by Analytics.log)

Format	Where	Event	Parameters
custom / house ads, link steps	HouseAdDirector.kt:110,188; ExitAdRunner.kt:143	ad (step show/ click/link_open)	

3. Log events (Firebase Analytics)

Event	Parameters	When it fires	Where (examples)
screen	screen, enter_time (ms), exit_time (ms), duration_sec, exit_by: button|next|back|home|recents|background, segment	One event per screen visit, sent when the visit ends (next screen shown, key press, or app goes background unless Screens.ownLeave). Names: splash, OB_1..OB_13/welcome/permission_hub/default_home (step key), onboarding_bottom_sheet, home, home_permission_sheet, settings, language, calculator, calc, list_<type>, detail, articles, article, uninstall_flow, uninstall_dialog, charging_event, package_event, recents_event, host_app_panel, launcher, app_drawer, suggested_panel, default_home_hint.	Screens.kt:18-20,85-100 (sender), SplashActivity.kt:119, OnboardingActivity.kt:256,559, HomeActivity.kt:64
click	screen (Screens.current), button, value (optional), segment	Any tap.	Analytics.kt:58-59, HomeContent.kt:112, SettingsActivity.kt:104, OnboardingActivity.kt:183,546
key_press	screen, key: back|home|recents, segment	System Back / Home / Recents; also sets the pending screen exit_by.	Analytics.kt:62-65, SystemKeys.kt:21-22, HomeActivity.kt:52, OnboardingActivity.kt:122,312
permission	screen, permission: camera|photos|notification|contacts|phone|overlay, result: show|allow|deny|already, segment	Each runtime permission asked from permissions.<screen>.	Analytics.kt:68-69, Permissions.kt:71,75,78
ad	screen, ad_place (RC place key, omitted for background preloads), format: banner|native|inter|rewarded|app_open|custom, step: request|loaded|failed|no_fill|show|show_failed|impression|click|close|timeout|link_open, error_code (failed/no_fill/show_failed), segment	Ad lifecycle for every renderer.	AdEvents.kt:29-41 (send/loadFailed), :78-91 (Listener: failed/no_fill/impression/click; FullScreen: show/show_failed/impression/click/close), BannerRenderer.kt:168,198,235, NativeAdRenderer.kt:136,146,153,181,184,202,418,423,426,440,494, NativeStripRenderer.kt:68,74,159
ad_impression	see paid[].params	AdMob OnPaidEventListener (revenue).	AdEvents.kt:44-56, see paid[]
default_home	step: dialog_show|settings_open|granted|settings_back|dialog_cancel, segment	Default-launcher request flow (launcher module).	Analytics.kt:72 via AppLauncherEvents.kt:51, launcher exts/Activity.kt:112, MainHubRequester.kt:90,104,112,126
app_update	step: shown|accepted|declined|failed|installed, value: force|flexible (on shown), segment	Play in-app update (base.app_update).	PlayPrompts.kt:35-43,53,68,69
rate_us	step: request|done|failed, segment	Play in-app review after prompts.rate_us.after_opens.	PlayPrompts.kt:134,138,141
uninstall	step: start|cancel|splash|language|step1|step2|progress|step3|reason|keep|fix|hidden|app_info, value: reason name (reason / fix), segment	Premium uninstall flow / dialog.	UninstallFlowActivity.kt:81,116,210,270,278,305,308, UninstallDialogActivity.kt:25,40
install_source	reason: no_referrer|no_gclid|fake_gclid|google_ip|gclid, segment	Once, when the install-referrer verdict is first decided.	InstallAttribution.kt:69 (reasons :119-125)
first_home	segment	Once, on the first Home visit after onboarding (pref first_home_pending).	HomeActivity.kt:45, PreferencesDataSource.kt:15-16

Event notes

• Central logger: ads/.../analytics/Analytics.kt:44-55 -> FirebaseAnalytics.logEvent. Allow-list Analytics.EVENTS (Analytics.kt:28) has 12 names; anything else is dropped with Log.e (the comment/log text still says '11 fixed events', stale).

• Every event carries segment = paid|organic (App.kt:64, from track).

• No first_* prefix rule exists: there is no automatic first-session renaming; first_home is a literal event.

• No user properties or user id are set (no setUserProperty / setUserId anywhere).

• No screen_view: screens use the custom 'screen' event; Firebase automatic screen_view still comes from the SDK's own activity tracking (not configured/disabled anywhere found).

• Param keys must match ^[a-z][a-z0-9_]{0,39}$; string values cut to 100 chars; Int->Long, Float->Double (EventRules, Analytics.kt:76-89).

• DebugEventLog.kt keeps a local copy of events in debug / forced-paid builds (notification list); it does not send anywhere.

Compared with the reference app

{
"summary": "None at event-name level. Reference 179-Body-Scanner-Home-01 ads/src/main/java/com/appkit/ads/{analytics/Analytics.kt, analytics/Screens.kt, host/AdEvents.kt} are byte-identical to ours after the package rename, and the same set of Analytics.log/step, permission, AdEvents.send step values and default_home/app_update/rate_us/uninstall/install_source/first_home calls exist in both.",
"differences": [
{
"what": "ad_impression with ad_placement 'inter_navigation'",
"reference": "179-Body-Scanner-Home-01/ads/src/main/java/com/appkit/ads/render/interstitial/NavigationInterstitial.kt:491 (presentGmsInterstitial)",
"ours": "Not present: our NavigationInterstitial has no show path (only preload, NavigationInterstitial.kt:255-275); every interstitial shows via ChainInterstitial (label inter_chain). Revenue is still logged, only the label differs."
},
{
"what": "ad_impression with ad_placement 'rewarded'",
"reference": "179-Body-Scanner-Home-01/ads/src/main/java/com/appkit/ads/render/RewardAdRenderer.kt:200 (reward-gated show)",
"ours": "Not present: rewarded shows only via presentIfLoaded (label rewarded_fallback, RewardAdRenderer.kt:101); ad_format 'rewarded' identical."
}
],
"not_compared": "Screen names and click button names are app-specific and were not diffed against the reference."
}

4. Remote Config wiring (every key)

Key	Parsed at	Used at	Status	Note
schema	-	-	not parsed	Version marker only; nothing reads it.
base.privacy_link	AppConfigParser.kt:72	SettingsActivity.kt:76	wired	
base.terms_link	AppConfigParser.kt:73	SettingsActivity.kt:77	wired	
base.store_link	AppConfigParser.kt:74	SettingsActivity.kt:170	wired	
base.update_link	AppConfigParser.kt:75	SettingsActivity.kt:74	wired	
base.app_update.show	AppConfigParser.kt:76	PlayPrompts.kt:47	wired	Parser default false (JSON true).
base.app_update.force	AppConfigParser.kt:77	PlayPrompts.kt:48	wired	
base.splash_wait_sec	AppConfigParser.kt:78	SplashActivity.kt:90,100	wired	
base.rc_sync_hours	AppConfigParser.kt:79	AppConfigStore.kt:79,117	wired	
base.attribution.gclid_min_length	AppConfigParser.kt:83	InstallAttribution.kt:130	wired	
base.attribution.fake_gclids	AppConfigParser.kt:84	InstallAttribution.kt:130	wired	
base.attribution.google_ip_check.on	AppConfigParser.kt:85	InstallAttribution.kt:63	wired	
base.attribution.google_ip_check.endpoint	AppConfigParser.kt:86	InstallAttribution.kt:97	wired	
base.attribution.google_ip_check.field	AppConfigParser.kt:87	InstallAttribution.kt:135	wired	
base.attribution.google_ip_check.match	AppConfigParser.kt:88	InstallAttribution.kt:136	wired	
base.exclusive_locations.on	AppConfigParser.kt:43,92	AppConfigParser.kt:44 (track override); ExclusiveLocations.kt:52 (lookup)	wired	
base.exclusive_locations.endpoint	AppConfigParser.kt:92	ExclusiveLocations.kt:53	wired	
base.exclusive_locations.list	AppConfigParser.kt:43,92	AppConfigParser.kt:44 (LocationRules.matches); ExclusiveLocations.kt:52	wired	
base.ad_network	AppConfigParser.kt:137	AdsConfigApplier.kt:31 -> BannerRenderer.kt:74, ChainInterstitial.kt:43, ResumeAdDirector.kt:75; ExitAdFlow.kt:111; BottomAdLoader.kt:44	wired	
base.ad_units.inter	AppConfigParser.kt:139	ExitAdRunner.kt:127; vault googleInter -> NavigationInterstitial.kt:246,256	wired	
base.ad_units.inter_splash	AppConfigParser.kt:140	ExitAdRunner.kt:125 (via AdUnits.interForSplash)	wired	Vault key googleS_Inter is written (AdsConfigApplier.kt:47) but never read; harmless.
base.ad_units.inter_back	AppConfigParser.kt:141	ExitAdRunner.kt:126; vault googleBackInter -> AdPreload.kt:21	wired	
base.ad_units.app_open	AppConfigParser.kt:142	vault googleAppopen -> ResumeAdDirector.kt:80	wired	
base.ad_units.native	AppConfigParser.kt:143	vault googleNative -> NativeStripRenderer.kt:53,174; AdUnitIds.kt:62 (NativeAdRenderer)	wired	

Key	Parsed at	Used at	Status	Note
base.ad_units.banner	AppConfigParser.kt:144	vault googleBanner -> BannerRenderer.kt:136	wired	
base.ad_units.rewarded	AppConfigParser.kt:145	vault googleRewarded -> RewardAdRenderer.kt:48	wired	
ads.on	AppConfigParser.kt:102	AdKit.kt:91 (adsOn, gates every Ads.* call); vault IsAdsON -> BannerRenderer.kt:53, NativeAdRenderer.kt:96, RewardAdRenderer.kt:44, AdPreload.kt:17	wired	
ads.loader_wait_sec	AppConfigParser.kt:103	AdsConfigApplier.kt:32 -> AdWait.kt:55 (BottomAdLoader.kt:55, ExitAdRunner.kt:112, RewardAdRenderer.kt:124)	wired	
ads.loader_on_miss	AppConfigParser.kt:104	ExitAdRunner.kt:121	wired	Parser default true (JSON false). Splash ignores it (always loads).
ads.preload.inter	AppConfigParser.kt:106	AdsConfigApplier.kt:41 -> AdLoadMode.kt:32 -> ChainInterstitial.kt:39,88; NavigationInterstitial.kt:244	wired	
ads.preload.native	AppConfigParser.kt:107	AdLoadMode -> NativeAdRenderer.kt:77,91; InterstitialBackupChain.kt:40	wired	
ads.preload.app_open	AppConfigParser.kt:108	AdLoadMode -> ResumeAdDirector.kt:65,195	wired	
ads.preload.rewarded	AppConfigParser.kt:109	AdLoadMode -> RewardAdRenderer.kt:43,112	wired	
ads.formats_on.inter	AppConfigParser.kt:112	ExitAdFlow.kt:113; vault InterAds -> NavigationInterstitial.kt:240	wired	
ads.formats_on.native	AppConfigParser.kt:113	ExitAdFlow.kt:116 (full_native); BottomAdLoader.kt:90; vault NativeAd -> NativeAdRenderer.kt:101,174, AdLifecycle.kt:31	wired	
ads.formats_on.banner	AppConfigParser.kt:114	BottomAdLoader.kt:88; vault BannerAdPresenter -> BannerRenderer.kt:53	wired	
ads.formats_on.native_banner	AppConfigParser.kt:115	BottomAdLoader.kt:89; vault NativeBannerPresenter -> NativeStripRenderer.kt:49,96	wired	
ads.formats_on.app_open	AppConfigParser.kt:116	ExitAdFlow.kt:115; vault AppopenAds -> ResumeAdDirector.kt:68	wired	
ads.formats_on.rewarded	AppConfigParser.kt:117	ExitAdFlow.kt:114; vault RewardedAds -> RewardAdRenderer.kt:44	wired	
ads.if_ad_not_ready	AppConfigParser.kt:119	ExitAdFlow.kt:84	wired	
ads.links	AppConfigParser.kt:120	AdsConfig.linksFor -> ExitAdFlow.kt:68; ExitAdFlow.kt:90; vault DirectLink -> AdSettingsVault.kt:51 (house-ad CTA)	wired	
ads.links_count	AppConfigParser.kt:121	AdsConfig.kt:32 -> ExitAdFlow.kt:68	wired	
ads.show_all	AppConfigParser.kt:122	AdsConfig.kt:37 -> ExitAdFlow.kt:77	wired	
ads.links_open_in	AppConfigParser.kt:123	AdsConfigApplier.kt:59 -> AdViewKit.kt:36 -> NavigationInterstitial.kt:112, AdViewKit.kt:54	wired	

Key	Parsed at	Used at	Status	Note
ads.custom_ads.on	AppConfigParser.kt:125	BottomAdLoader.kt:43; ExitAdFlow.kt:110; vault IsCustomADS -> HouseAdDirector.kt:53, InterstitialBackupChain.kt:53	wired	
ads.custom_ads.list	AppConfigParser.kt:126-133	vault CUSTOM_ADS -> HouseAdDirector.kt:61; BottomAdLoader.kt:43	wired	List empty in JSON; item keys icon,title,description,bannerImage,buttonText.
ads.colors.{light,dark}.{button,button_text,background,text}	AppConfigParser.kt:136,162-167	AdsConfigApplier.kt:73-80 -> NativeAdRenderer.kt:540-543,678-681	wired	Only #RRGGBB accepted; written before each slot/exit (BottomAdLoader.kt:41, ExitAdRunner.kt:57).
ads.<group>.<place>.bottom_ad.{on,type}	AppConfigParser.kt:170,176-178	Ads.kt:48 -> BottomAdLoader.kt:39,104; ScreenAd.kt:36 (type picks scroll vs pinned container)	wired	
ads.<group>.<place>.bottom_ad.on_fail	AppConfigParser.kt:179	BottomAdLoader.kt:70,112	wired	Only applies when bottom_ad.type is a banner and on_fail is not a banner.
ads.<group>.<place>.list_ad.{on,type}	AppConfigParser.kt:171	Ads.kt:51,55; ScreenAd.kt:23; HomeContent.kt:107; OnboardingActivity.kt:539-545	wired	
ads.<group>.<place>.list_ad_after / list_ad_every	AppConfigParser.kt:173 (raw options)	ScreenAd.kt:24 (Activity.listAd)	wired	Only for places read through listAd(): screens.calculator, settings, list_*, list_*_items, tips, diamond_tips. Missing list_ad_every = 0 = once.
ads.<group>.<place>.list_ad_position	AppConfigParser.kt:173 (raw options)	HomeContent.kt:101	wired	Only screens.home and launcher.host_app_panel (HomeContent).
ads.<group>.<place>.exit_ad.{on,ads}	AppConfigParser.kt:182-184	ExitAdRunner.kt:43,49 -> ExitAdFlow.kt:130,74-78	wired	
ads.<group>.<place>.exit_ad.links_count	AppConfigParser.kt:185	AdsConfig.kt:32 -> ExitAdFlow.kt:68	wired	
ads.<group>.<place>.exit_ad.every	AppConfigParser.kt:188	ExitAdFlow.kt:28,130	wired	In JSON only screens.list (=1).
ads.<group>.<place>.gap_sec	AppConfigParser.kt:173 (raw)	ExitAdRunner.kt:52 (non-event places); ChargeEventWatcher.kt:88; PackageEventWatcher.kt:116; RecentAdWatcher.kt:115	wired	
ads.<group>.<place>.on	AppConfigParser.kt:173 (raw)	Ads.kt:171 placeOn -> UnlockAdWatcher.kt:99, ChargeEventWatcher.kt:87, PackageEventWatcher.kt:57,115, RecentAdWatcher.kt:150	wired	
ads.screens.splash.exit_ad.*	AppConfigParser.kt:147-152	SplashActivity.kt:113 (Ads.SPLASH)	wired	Off in track_b.
ads.screens.home.{bottom_ad,list_ad,list_ad_position,exit_ad}	AppConfigParser.kt:147-152	HomeContent.kt:97-107 (slots), HomeContent.kt:113 (exit on every Home destination)	wired	
ads.screens.settings.{exit_ad,list_ad}	AppConfigParser.kt:147-152	SettingsActivity.kt:45 (exit on close), :62 (list_ad under Language row), :88 (bindScreenAd: bottom_ad, track_a only)	wired	
ads.screens.list.{bottom_ad,exit_ad.every}	AppConfigParser.kt:147-152	ListActivity.kt:37 (exit on item tap), :44 (bottom)	wired	
ads.screens.detail.bottom_ad	AppConfigParser.kt:147-152	DetailActivity.kt:68	wired	
ads.screens.articles.bottom_ad	AppConfigParser.kt:147-152	ArticlesActivity.kt:48	wired	

Key	Parsed at	Used at	Status	Note
ads.screens.article.bottom_ad	AppConfigParser.kt:147-152	ArticlesActivity.kt:83	wired	
ads.screens.calculator.{bottom_ad,list_ad,list_ad_after}	AppConfigParser.kt:147-152	CalculatorActivity.kt:33 (listAd), :35 (exit, no exit_ad in JSON -> passes through), :37 (bottom)	wired	
ads.screens.calc.{bottom_ad,list_ad}	AppConfigParser.kt:147-152	CalculatorActivity.kt:71 (Ads.list above Calculate), :72 (bottom)	wired	
ads.screens.language.{bottom_ad,exit_ad}	AppConfigParser.kt:147-152	LanguageActivity.kt:39 (exit), :45 (bottom)	wired	
ads.screens.list_{characters,bundles,pets,vehicles,emotes,weapons,maps}.list_ad*	AppConfigParser.kt:147-152	ListActivity.kt:34-35 (place = screens.list_<CatalogType>)	wired	
ads.screens.list_{emotes,weapons,maps}_items.list_ad*	AppConfigParser.kt:147-152	ListActivity.kt:34 (suffix _items when inside a category)	wired	list_characters_items etc. not in JSON -> no list ad inside those categories (only reached when a section has >1 group; not verified which do).
ads.screens.tips.list_ad*	AppConfigParser.kt:147-152	ArticlesActivity.kt:46	wired	
ads.screens.diamond_tips.list_ad*	AppConfigParser.kt:147-152	ArticlesActivity.kt:46	wired	
ads.screens.diamond_article.bottom_ad	AppConfigParser.kt:147-152	ArticlesActivity.kt:82	wired	
ads.onboarding.OB_1..OB_13.{bottom_ad,list_ad,exit_ad}	AppConfigParser.kt:147-152	OnboardingActivity.kt:539-545 (slots), :206-207 (exit on advance)	wired	
ads.onboarding.default_home.{bottom_ad,exit_ad}	AppConfigParser.kt:147-152	OnboardingActivity.kt:542 (bottom), :449 (exit after default-home granted), :206	wired	
ads.onboarding.welcome.*	AppConfigParser.kt:147-152	OnboardingActivity.kt:539-545,206	wired	Code path wired, but 'welcome' is not in track_b onboarding.order, so never reached with this RC.
ads.onboarding.permission_hub.*	AppConfigParser.kt:147-152	OnboardingActivity.kt:539-545,206	wired	Same: 'permission_hub' not in onboarding.order -> unreachable with this RC.
ads.launcher.app_resume.exit_ad	AppConfigParser.kt:147-152	Ads.kt:97 (AppResumeWatcher.kt:31)	wired	Off in track_b.
ads.launcher.back_button.{exit_ad,during_onboarding}	AppConfigParser.kt:147-152	HomeActivity.kt:53 (Ads.backButton); Ads.kt:65 -> OnboardingActivity.kt:206,231; AdsConfigApplier.kt:60 (IsBack preload)	wired	
ads.launcher.home_button.{exit_ad,during_onboarding}	AppConfigParser.kt:147-152	Ads.kt:65 -> OnboardingActivity.kt:206,231-232	wired	Only runs during onboarding (needs during_onboarding true). No post-onboarding Home-press call site found (unlike recents_button RecentsPolicy.kt:149 / back_button HomeActivity.kt:53). Unsure whether intended.
ads.launcher.recents_button.{exit_ad,during_onboarding}	AppConfigParser.kt:147-152	RecentsPolicy.kt:149 (after onboarding, we are default home); OnboardingActivity.kt:206,231	wired	
ads.launcher.app_exit.exit_ad	AppConfigParser.kt:147-152	Ads.kt:94,126 (AppLauncherEvents.kt:26)	wired	
ads.launcher.app_open.exit_ad	AppConfigParser.kt:147-152	Ads.kt:120 (AppLauncherEvents.kt:24)	wired	
ads.launcher.host_app_panel.{bottom_ad,list_ad,list_ad_position,exit_ad}	AppConfigParser.kt:147-152	HostPanelFragment.kt:38 -> HomeContent.kt:97-107; Ads.kt:115 (exit on panel open, AppLauncherEvents.kt:20)	wired	launcher.panels.host_app=false in track_b -> panel disabled (HubPromosSwitches.kt:29), so these ads are unreachable with this RC. Note: taps inside the panel run screens.home exit_ad (HomeContent.kt:113), not host_app_panel's.
ads.launcher.suggested_panel.{bottom_ad,list_ad,exit_ad}	AppConfigParser.kt:147-152	AppLauncherEvents.kt:20,33,34	wired	panels.suggested_panel=false and all slots off -> unreachable with this RC.

Key	Parsed at	Used at	Status	Note
ads.launcher.app_drawer.{exit_ad,bottom_ad}	AppConfigParser.kt:147-152	Ads.kt:117 (AppLauncherEvents.kt:22); AppLauncherEvents.kt:35	wired	
ads.launcher.app_drawer.bottom_ad_row	AppConfigParser.kt:230	HubPromosSwitches.kt:34 (LauncherSettings.drawerAdRow)	wired	
ads.launcher.drawer_sponsored_apps.{on,every,list[logo, landing_url,label]}	AppConfigParser.kt:224-228	PromotedTiles.kt:23-32	wired	Read into LauncherSettings.sponsoredApps (also lands in AdsConfig.places as an empty place; harmless). Items with blank landing_url dropped.
ads.launcher.sponsored_app_open.exit_ad	AppConfigParser.kt:147-152	Ads.kt:129 (AppLauncherEvents.kt:28)	wired	
ads.launcher.sponsored_app_exit.exit_ad	AppConfigParser.kt:147-152	Ads.kt:95,134 (AppLauncherEvents.kt:30)	wired	
ads.launcher.phone_unlock.{on,exit_ad,start_after_hours ,gap_minutes,max_per_day}	AppConfigParser.kt:147-152,17 3	UnlockAdWatcher.kt:94,99-110	wired	
ads.launcher.charger_{connected,disconnected}.{on,bott om_ad,exit_ad,gap_sec}	AppConfigParser.kt:147-152,17 3	ChargeEventWatcher.kt:87-88; ChargingStatusActivity.kt:149 (bottom), :156 (exit on close)	wired	on=false in track_b.
ads.launcher.app_{installed,uninstalled}.{on,bottom_ad,e xit_ad,gap_sec}	AppConfigParser.kt:147-152,17 3	PackageEventWatcher.kt:57,115-116; PackageResultActivity.kt:289,298	wired	on=false in track_b.
ads.launcher.play_store_return.{on,window_sec,show_ad _screen,bottom_ad,exit_ad,gap_sec}	AppConfigParser.kt:147-152,17 3	RecentAdWatcher.kt:111-115,150-153; RecentAdActivity.kt:27,47	wired	Code default window_sec 180 (JSON 6000).
ads.uninstall_flow.splash.exit_ad	AppConfigParser.kt:147-152	UninstallFlowActivity.kt:195 (leaveWhenFree) -> :257/243	wired	
ads.uninstall_flow.{step1,step2,step3,progress,language} .bottom_ad	AppConfigParser.kt:147-152	UninstallFlowActivity.kt:124 (place = uninstall_flow.<screen>)	wired	
ads.uninstall_flow.language.exit_ad	AppConfigParser.kt:147-152	UninstallFlowActivity.kt:247-257 (pickLanguage -> leave)	wired	Other screens call Ads.exit too but have no exit_ad in JSON -> pass through.
onboarding.order	AppConfigParser.kt:192	OnboardingRules.kt:13	wired	default_home removed when launcher off or already default home.
onboarding.skip_button	AppConfigParser.kt:193	OnboardingViewModel.kt:36 -> OnboardingActivity.kt:531	wired	
onboarding.default_home.skip_button	AppConfigParser.kt:194	OnboardingViewModel.kt:37 -> OnboardingActivity.kt:531	wired	
onboarding.default_home.auto_open_settings_sec	AppConfigParser.kt:195	OnboardingActivity.kt:435	wired	
permissions.permission_hub	AppConfigParser.kt:198-201	Permissions.kt:54 via OnboardingActivity.kt:183	wired	Unreachable with this RC (permission_hub not in onboarding.order). Every onboarding step also asks permissions.<step key> (OnboardingActivity.kt:534) - none configured.
permissions.home	AppConfigParser.kt:198-201	HomePermissionSheet.kt:35-36; Permissions.kt:54	wired	
permissions.home_sheet	AppConfigParser.kt:61	HomePermissionSheet.kt:35	wired	
launcher.on	AppConfigParser.kt:206	App.kt:84-85; OnboardingRules.kt:13; SettingsActivity.kt:52; OnboardingActivity.kt:459	wired	
launcher.os_style	AppConfigParser.kt:207	LobbyPersonaApplier.kt:68	wired	

Key	Parsed at	Used at	Status	Note
launcher.default_home_banner	AppConfigParser.kt:208	SettingsActivity.kt:52	wired	
launcher.our_icon.in_drawer	AppConfigParser.kt:209	AppLauncherHost.kt:22 -> BoardPane.kt:229, GadgetsStagePane.kt:137	wired	
launcher.our_icon.on_home	AppConfigParser.kt:210	AppLauncherHost.kt:22 -> LobbyLookerMatrix.kt:296	wired	
launcher.panels.host_app	AppConfigParser.kt:211	HubPromosSwitches.kt:29 -> XHubStage.kt:169,323	wired	
launcher.panels.suggested_panel	AppConfigParser.kt:212	HubPromosSwitches.kt:28 -> XHubStage.kt:166	wired	
launcher.swipe_guide.on	AppConfigParser.kt:215	HubPromosSwitches.kt:19	wired	
launcher.swipe_guide.{swipe_right_host_app,swipe_left_suggested_panel,swipe_up_drawer}	AppConfigParser.kt:216-218	HubPromosSwitches.kt:21-23	wired	
launcher.time_widget	AppConfigParser.kt:221	LobbyWidgetPositioner.kt:238 -> :50,53,115	wired	
launcher.search_widget	AppConfigParser.kt:222	LobbyWidgetPositioner.kt:243 -> :71,127	wired	Search row itself is always shown (LobbyWidgetPositioner.kt:241 shortcut); key picks google/chrome only.
launcher.system_keys.home	AppConfigParser.kt:241	OnboardingRules.kt:31; AppLauncherHost.kt:42	wired	
launcher.system_keys.back	AppConfigParser.kt:242	OnboardingRules.kt:31	wired	
launcher.system_keys.recents	AppConfigParser.kt:243	OnboardingRules.kt:31; RecentsPolicy.kt:154	wired	
launcher.system_keys.bottom_sheet.{on,keys}	AppConfigParser.kt:239,244-245	OnboardingRules.kt:30; RecentsPolicy.kt:154	wired	
uninstall_flow.flow	AppConfigParser.kt:250	AppLauncherHost.kt:24; AppLauncherEvents.kt:40; UninstallFlowActivity.kt:70; UninstallDialogActivity.kt:24	wired	
uninstall_flow.system_dialog	AppConfigParser.kt:251	AppLauncherEvents.kt:41; UninstallDialogActivity.kt:24	wired	
uninstall_flow.splash_ms	AppConfigParser.kt:252	UninstallFlowActivity.kt:126	wired	
uninstall_flow.progress_ms	AppConfigParser.kt:253	UninstallFlowActivity.kt:170 (clamped 500..20000)	wired	
prompts.rate_us.after_opens	AppConfigParser.kt:66	PlayPrompts.kt:146	wired	
prompts.rate_us.retry_after_hours	AppConfigParser.kt:66	PlayPrompts.kt:146	wired	
countries.<CC[,CC]>.* (e.g. IN,PK.permissions.*)	AppConfigParser.kt:47-52 (deep merge into the track)	Permissions.kt:54 / HomePermissionSheet.kt:36 (for the permissions overlay in JSON)	wired	Country from SIM/network/locale (Resolvers.kt:19). Any track key can be overlaid.

Read by the parser but not in the JSON (optional keys)

Key	Note
ads.<group>.<place>.exit_ad.links	Per-block link list override; not in JSON.

Key	Note
ads.<group>.<place>.exit_ad.show_all	Per-block override of ads.show_all; not in JSON.
ads.<group>.<place>.preload	Per-place inter preload override; not in JSON.
ads.<group>.<place>.exit_ad.every (on places other than screens.list)	Default 1.
ads.<group>.<place>.list_ad_every (places without it)	Missing = 0 = one ad only.
ads.launcher.<event>.window_sec/start_after_hours/gap_minutes/max_per_day defaults	Code defaults when missing.
bottom_ad/list_ad type values banner_inline, banner_collapsible, native_full	Accepted values not used in JSON.
exit_ad.ads value 'custom'	Not used in JSON.
launcher.system_keys.<key> values 'reset' and '' (device default)	Not used in JSON.
permissions values contacts / phone / overlay	Accepted, not in JSON.
uninstall_flow.flow 'normal'	Any value other than 'premium' = normal.
Remote Config params home_profile_rules / home_profile_probes (separate RC keys, not inside app_config)	Read alongside app_config, injected into LauncherSettings.
Vault per-placement overrides (DirectLink, link_open_in, googleInter, googleFullNative per place)	Engine can read per-placement values, but AdsConfigApplier only writes 'preload' into placements, so these overrides are always empty. Dead capability, not an RC key.

5. In-app update

• Library: com.google.android.play:app-update(-ktx) 2.1.0 - same version as the reference.

• RC: base.app_update.show (offer an update) and base.app_update.force (mandatory). Live RC: show = true, force = false.

• Flexible (show): offered once per app run (not again on the next screen), downloads in the background, installs on the next return (completeUpdate).

• Forced (force): full-screen update on every resume; backing out closes the app - now the same as the reference (the extra 'Update required' dialog was removed).

• Analytics: app_update event with step shown (flexible/force), accepted, declined, failed, installed.

• Version: versionCode 1, versionName 1.0.0 (reference: 1 / 1.0.0). Play only offers an update when its versionCode is HIGHER than the installed one, so raise versionCode by 1 for every Play upload (2, 3, ...). It can be tested only with an app installed from Play (internal testing / internal app sharing), not with a local debug build.